

National University of Computer and Emerging Sciences

FAST-NUCES, Karachi Campus

Project Proposal

Real-Time Messaging Application

Flutter with SQL Backend

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Project Title

Real-Time Messaging Application using Flutter and PostgreSQL

Objective

To develop a cross-platform mobile chat application that demonstrates the practical use of a relational SQL database (PostgreSQL) for storing and querying user and message data in real time.

Problem Statement

Most modern chat applications rely on NoSQL databases, which sacrifice structure and relational integrity for scalability. This project aims to demonstrate that a well-designed SQL schema can effectively support real-time messaging with proper normalization, foreign key constraints, and efficient querying.

Proposed Solution

A Flutter-based messaging application will be developed with the following components:

- **Frontend:** Flutter (Dart) for cross-platform UI on Android and Web.
- **Authentication:** Supabase Auth for user registration and session management.
- **Database:** PostgreSQL hosted on Supabase with two core tables — **users** and **messages**.
- **Realtime:** Supabase stream subscriptions for live message delivery.

Database Schema (Proposed)

Table	Column	Type
users	id	UUID (PK)
	email	TEXT
messages	id	UUID (PK)
	sender_id	UUID (FK)
	receiver_id	UUID (FK)
	chat_room_id	TEXT
	message	TEXT
	timestamp	TIMESTAMPTZ

Expected Outcomes

- A functional real-time chat application.
- A normalized SQL schema demonstrating relational database concepts.
- Demonstration of SQL integration in a modern mobile application.

Tools and Technologies

Flutter, Dart, Supabase, PostgreSQL, GitHub Actions (CI/CD).